Coca Is Not Cocaine

Three Lexical-Cultural Collision Modes in Open-Weight LLMs, Probed in Catalan

Xavier Vinaixa Roselló

Sorensen AI, Barcelona, Catalonia

ORCID 0009-0005-2769-9215 | xavi@sorensen.ai

2026 | *Preprint — not peer-reviewed*

DOI: 10.13140/RG.2.2.10368.39682

Abstract. Culturally-marked Catalan concepts collide with the Anglo-Saxon imaginary in at least three structurally distinct ways, and naive cultural-bias probes conflate them into a single number. A *gegant* is not a fairy-tale *giant* (it is a festive effigy), a *coca* is not cocaine (it is a flatbread), and a *pinya* at the base of a *castell* is neither a pinecone nor a pineapple. We extend the CAMeL framework (Naous et al., 2024) to a panel of **eleven open-weight base LLMs** through contextualized hidden-state geometry — mean-pooled, L2-normalized vectors at a best layer chosen by KNN purity over anchors — and stratify the bias ratio $\text{pull}_{\text{anglo}}/(\text{pull}_{\text{anglo}} + \text{pull}_{\text{catalan}})$ along an `ambiguity_type` axis. This separates (i) cross-lingual cognate drag (*gegant*, *castell*, *coca*), (ii) purely domain-specific representation gaps (*correfoc*, *casteller*, *romesco*), and (iii) intra-Catalan polysemy (*pinya*, *tió*, *folre*, *manilles*). The panel pairs four general-purpose families (Gemma, Qwen, Mistral, DeepSeek) with a Catalan-aware control tier (Salamandra/ALIA from the Barcelona Supercomputing Center). The central result is that all five families exhibit the **same collision pattern** in the absence of context, which locates the cause of the collision in the *contextualized-embedding pipeline* rather than in the training-data mixture. A pre-registered title test confirms that the literal reading —*coca* adjacent to *cocaine*, *Coca-Cola*, *coke*— is a property of latent space in ten of eleven models. The practical intervention is complementary to any pretraining strategy: supply Catalan context in the inference-time prompt.

Keywords: cultural bias, LLM, Catalan, contextualized embeddings, low-resource languages, Salamandra, ALIA.

1 Introduction

Large language models (LLMs) are trained on corpora massively dominated by English. A known consequence is cultural misalignment: models systematically reproduce the Western — and especially Anglo-Saxon — imaginary when asked to reason about non-anglophone societies (Tao et al., 2024; Wang et al., 2023; Ada Lovelace Institute, 2025). Most existing cultural-bias probes measure this with a single aggregate number (Naous et al., 2024; Li et al., 2023), which hides the fact that a culturally-marked concept can fail in qualitatively different ways.

This work takes Catalan as a limiting case and asks a concrete, measurable question: *when a base LLM reads a culturally-marked Catalan word, where does it place it in latent space?* A *coca* is a sweet or savoury flatbread typical of Catalan cuisine; in English, *coca* points to the plant, the drug, or the soft-drink brand. A *gegant* is a festive papier-mâché figure that parades through town squares; its English cognate *giant* lives in a fairy-tale universe. And a *pinya* can be a pinecone, a pineapple, or the human base of a *castell* (human tower), depending on context.

We argue that these three examples illustrate three *structurally distinct failure modes*, which we call **F1**, **F2** and **F3** (Section 3), and that they deserve different diagnostics — and interventions. The central contribution is methodological: we show that stratifying the bias metric by ambiguity type reveals stories that the aggregate metric hides.

Contributions.

1. A **three-mode taxonomy** (**F1**/**F2**/**F3**) of lexical-cultural collisions, with a hand-curated 56-concept Catalan lexicon stratified by ambiguity type.
2. A **geometric adaptation of CAMeL** (Naous et al., 2024) from fill-in-the-blank to contextualized-embedding geometry, with bootstrap confidence intervals per (concept, frame).
3. An **eleven-model base panel** pairing four general-purpose families with a **causal Catalan-aware control** (Salamandra/ALIA from the BSC), which lets us distinguish the "the problem is the data" hypothesis from the "the problem is the pipeline" hypothesis.
4. A **pre-registered title test** that literally validates the *coca* ↔ *cocaine* collision in latent space.

2 Related work

Direct methodological inspiration. The reference framework is **CAMeL** (Naous et al., 2024), which measures cultural bias with fill-in-the-blank probes in Arabic. This project takes it (i) from fill-in-the-blank to contextualized-embedding geometry, (ii) from Arabic to Catalan, and (iii) adds the **F1**/**F2**/**F3** taxonomy that separates the failure modes CAMeL aggregates into a single metric.

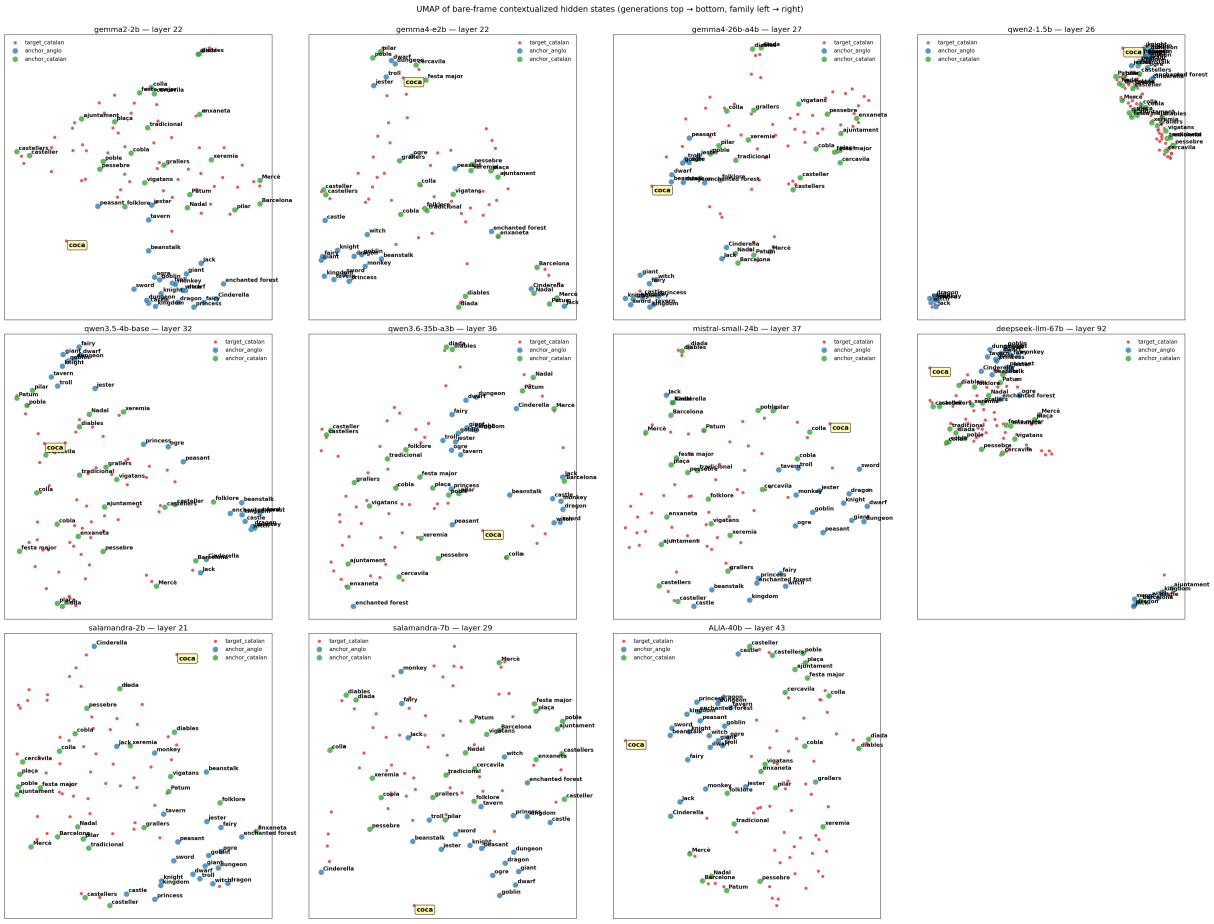


Figure 1: UMAP of the contextualized hidden states in the `bare` frame at each model's best layer, in a grid ordered by family (Gemma → Qwen → Mistral → DeepSeek → Salamandra → ALIA) and then by size. *coca* is highlighted with a yellow callout in every panel: following it across models shows the title concept's journey through eleven different latent spaces.

The cultural-bias trunk. Tao et al. (2024) show systematic cultural misalignment in LLM outputs about non-Western societies — the macro "why it matters" behind our micro-geometric work. Li et al. (2023) (GeoMLAMA) confirm that the problem cuts across probe modalities. Rystrøm et al. (2025) argue that training on more languages does not imply capturing more cultures (*multilingual* ≠ *multicultural*): this is exactly the "pipeline, not data-mix" reading that the Salamandra/ALIA family confirms empirically here. Wang et al. (2023) document Anglo-American cultural dominance in the training mix, the direct motivation for using Catalan as a limiting case.

Classic embedding-geometry tradition. Bolukbasi et al. (2016) is the founding work on bias probes via cosine proximity to anchor sets. Our bias_ratio = $\text{pull}_{\text{anglo}}/(\text{pull}_{\text{anglo}} + \text{pull}_{\text{catalan}})$ is its contextualized extension, applied to a cultural collision rather than a gender collision.

Methodological constraint. Dettmers et al. (2022) show that quantization reorganizes representation geometry. This grounds the "bf16 only, no fp8 / int8 / NVFP4" rule applied across the whole panel (Section 4.3).

3 The F1/F2/F3 taxonomy

A culturally-marked target word c in a low-resource language can fail in three structurally distinct ways when a probe asks where c sits in a multilingual base LLM's latent space. The 56-concept lexicon (`data/concepts_ca.csv`) is stratified by an `ambiguity_type` column that tags each target with exactly one of the codes in Table 1.

Why each stratum needs a different intervention.

- **F1** confounds *anglification* with *cognate semantics*: when *gegant* lands near *giant*, that is partly because they share an English referent and partly because English-dominant pretraining drags the Catalan word toward its cognate. A pull is observed but its cause is ambiguous by construction.
- **F2** is the cleanest evidence stratum. *Correfoc* has no English cognate; if it lands inside the Anglo fairy-tale cluster the pull is structural — there is no lexical shortcut to fall onto.
- **F3** can produce a high pull *without* anglification: if *pinya* lands near *pinecone*, the model is just reflecting that the mundane Catalan sense (the fruit) is more frequent in the training mix than the festive one (the

Code	Type	n	Failure mode / examples
F1	`cross_lingual_bias`	10	Direct English cognate, different universe. *gegant*↔giant, *castell*↔castle, *drac*↔dragon, *coca*↔coca, *Pasqua*↔Easter.
F2	`domain_specific`	29	No English cognate — pure domain depth. *correfoc*, *casteller*, *cercavila*, *panellet*, *grallers*, *pa amb tomàquet*, *samfaina*, *calçotada*, *romesco*.
F3	`internal_polysemy`	17	Mundane Catalan sense dominates the festive one. *pinya* (pinecone ‖ castell base), *tió* (log ‖ Christmas log), *folre* (lining ‖ castell padding), *manilles* (handcuffs ‖ castell grip).

Table 1: The three-mode taxonomy of lexical-cultural collisions.

`frame_id`	Sentence
`bare`	*c* alone, no context.
`anglo_en`	*Once upon a time, there was a c who lived in the enchanted forest.*
`catalan_frame`	*A la festa major del poble, els c van sortir a la plaça acompanyats dels grallers.*
`neutral_ca`	*El c és un element important.*

Table 2: The four probe frames.

castell base). The fix is not "teach the model that *pinya* ≠ *giant*"; it is "teach the model that *pinya* has a second sense that is not a pinecone".

F1 audit. **F1** is deliberately kept *tight* (10 concepts) and conservative. A concept enters **F1** only if it has a direct English cognate **and** that cognate lives in a clearly different cultural universe. The audit removed three concepts (*mare*, *port*, *vall*) whose cognates were culturally co-extensive with their Catalan senses, and moved *terra* from **F1** to **F3** because its interesting case (*la terra* as political territory vs. *terra* as soil) is intra-Catalan polysemy rather than cross-lingual collision.

4 Methodology

4.1 Probe-frame design

For each target concept c the pipeline emits four probes (Table 2). We deliberately leave the Catalan lemma verbatim in `anglo_en` even when c has a direct English cognate: substituting *gegant*→*giant* would measure *giant*'s geometry, not the cross-lingual pull on *gegant*.

4.2 Anchor sets

Three hand-curated clusters are used: Anglo fairy-tale / medieval vocabulary (22 entries, the dominant cluster), Catalan folkloric vocabulary (23 entries, the target cluster), and a drug / brand / beverage cluster (13 entries, used *only* for the title test). The Anglo cluster is subdivided into `fairy_tale`, `medieval` and `character` so the per-concept analysis can pinpoint *which* English neighbourhood catches each focus concept when it drifts.

4.3 Hidden-state extraction

For each probe (c, f) and each requested layer L: (1) tokenize the sentence with the model's fast tokenizer (`return_offsets_mapping=True`); (2) map the character span of c to its token span via `offset_mapping`; (3) forward pass with `output_hidden_states=True`, `use_cache=False`, no gradients; (4) slice `hidden_states[L][0, t_start:t_end, :]`, mean-pool over the subword positions, and L2-normalize.

All extractions use `torch_dtype=bfloat16` and `device_map="auto"`. **Quantization is forbidden**: fp8 / int8 / NVFP4 reorganizes the representation geometry the metrics measure (Dettmers et al., 2022).

4.4 Metrics

`bias_ratio`. For each (c, f):

$$\text{pull}_{\text{anglo}}(c,f) = \operatorname*{mean}_{a\in\text{ANGLO}} \cos\big(v(c,f),\, v(a,\text{bare})\big), \quad (1)$$

$$\text{pull}_{\text{cat}}(c,f) = \operatorname*{mean}_{a\in\text{CAT}} \cos\big(v(c,f),\, v(a,\text{bare})\big), \quad (2)$$

$$\text{bias_ratio}(c,f) = \frac{\text{pull}_{\text{anglo}}}{\text{pull}_{\text{anglo}} + \text{pull}_{\text{cat}}}, \quad (3)$$

with 95 % CIs from 1000 bootstrap resamples over the anchor sets. Surface-string self-matches are excluded.

`top10_catalan_frac` (complementary). Raw cosines saturate near 0.5 on both clusters at deep layers, which compresses the dynamic range of `bias_ratio`. We therefore also report a rank-based metric: of the 10 anchors nearest to $v(c, f)$, what fraction belong to the Catalan cluster? `1.0` = the whole top-10 KNN is Catalan folkloric; `0.0` = entirely Anglo fairy-tale.

Frame differential Δ. Writing br for the bias ratio,

$$\Delta(c) = \text{br}(c, \text{anglo_en}) - \text{br}(c, \text{cat_frame}). \quad (4)$$

Δ is the headline diagnostic. A large positive Δ means the model holds retrievable Catalan geometry that an explicit Catalan frame unlocks; $\Delta \approx 0$ means the representation is structurally Anglo-anchored regardless of framing. The CI on Δ uses a **paired** bootstrap: each iteration reuses the same anchor index for both frames before differencing, cancelling the shared noise and materially tightening the interval.

4.5 Layer selection

For each model we sweep three layers at 25 / 60 / 90 % depth and pick the best by "deepest layer whose anchor-KNN purity is within 0.15 of the model's per-layer max". Early layers win raw purity by surface morphology (English vs. Catalan orthography), which is not what we want to probe; the tolerance keeps us in the semantic regime. The rule shares its anchor pool with the metrics, so it is mildly circular; we mitigate by reporting every metric across all three committed layers and adding the per-concept paired-bootstrap CI.

5 The model panel

The panel comprises eleven base (pre-RLHF) models in three tiers: four general-purpose families at small and large scale, plus a Catalan-aware control tier of three BSC models (Table 3). Only `-base` variants are probed: instruction tuning (RLHF/SFT) collapses the representation toward "respond to the user", masking the pretraining-bias signal the probe is designed to read.

6 Pre-registered title test

The project's title makes a literal claim (*coca is not cocaine*), not just an **F2** metaphor (*coca* drifting toward fairy-tale vocabulary). To test the literal claim we add the noise cluster (`anchors_noise.csv`: 13 drug / brand / beverage words) and count how many land in the top-10 KNN of *coca* per (model, frame). The pre-specified decision rule — committed before the cloud extraction — was: *falsified* if $\max n_{\text{noise}} < 2$ in every cell (reframe the title); *partially supported* if $n_{\text{noise}} \geq 2$ *only* under `anglo_en` (prompt artifact); **strongly supported (structural)** if $n_{\text{noise}} \geq 2$ in `bare`, `catalan_frame` *or* `neutral_ca` (keep the title; a property of latent space). The extraction returns the **strongly supported** branch on ten of eleven models (Section 7.3).

7 Results

7.1 Bias ratio stratified by ambiguity type

Table 4 reports the median `bias_ratio` pooled across ambiguity types (n = 56 concepts). Gemma 4 26B has the **deepest native drop** (0.483 `bare` → 0.370 `catalan_frame`) and the **lowest absolute Anglo pull** under Catalan framing of any model. The biggest Qwen (3.6 35B) is noticeably *less* responsive to Catalan framing than the smaller Qwen 3.5 4B: more parameters do not translate into more recoverable Catalan geometry. DeepSeek 67B-base is the **least native-frame-responsive of the entire panel** (drop 0.030, comparable to Salamandra-7B despite having no targeted Catalan training).

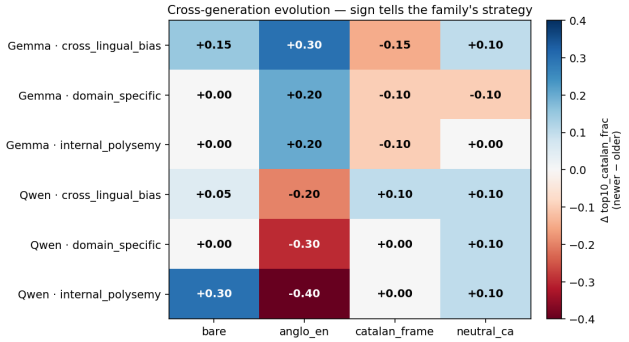


Figure 2: top10_catalan_frac(newer) − top10_catalan_frac(older). Blue = the newer model places the concept more Catalan; red = it regressed. The two families move in opposite directions.

7.2 The two families moved in opposite directions

We probed both the current and the previous generation per family: Gemma 2 2B → Gemma 4 E2B and Qwen2 1.5B → Qwen 3.5 4B. The delta is top10_catalan_frac(newer) − top10_catalan_frac(older) (Figure 2).

- **Google (Gemma 2 → 4): more context-invariant.** The English-frame floor rises (+0.20 on `cross_lingual_bias`) but the Catalan-frame ceiling drops (−0.10). The newer Gemma is harder to push around but cannot reach perfect Catalan placement under explicit native context.
- **Alibaba (Qwen 2 → 3.5): more context-amplified.** Catalan-frame placement climbs to a perfect `1.0` across every stratum, but English-frame placement collapses by ∼0.30 on `cross_lingual_bias` and ∼0.40 on `internal_polysemy`.

There is no global "best LLM for Catalan": the right model depends on whether the deployment can guarantee Catalan framing (then Qwen) or must survive mixed / untrusted surrounding text (then Gemma). The relationship is an axis, not a ranking.

7.3 "Coca is not cocaine" — the title case

Under `anglo_en` —*"Once upon a time, there was a coca who lived in the enchanted forest"*— **Salamandra 7B places 9 of 10** nearest anchors in the noise cluster, ALIA-40B places 8, and Mistral Small 24B / Qwen 3.5 4B / Salamandra 2B place 7 each. Two outliers: Gemma 4 E2B and DeepSeek 67B-base place **zero** noise anchors — DeepSeek's top-3 is *enchanted forest, fairy, princess*, drifting cleanly into the fairy-tale cluster instead (pure **F2**).

Under `bare` (no carrier sentence) the effect persists: 7 noise hits on ALIA-40B, Mistral Small 24B and Salamandra 2B; 6 on DeepSeek 67B, Gemma 2, Qwen 2 and Salamandra 7B; and 0 only on Gemma 4 E2B. This is what *structural* means in the decision rule: the collision

Model	Hugging Face ID	layer sweep	best layer
Small tier + previous generation (16 GB GPU)			
Gemma 2 2B base	`google/gemma-2-2b`	6, 15, 22	22
Gemma 4 E2B base	`google/gemma-4-E2B`	6, 15, 22	22
Qwen2 1.5B base	`Qwen/Qwen2-1.5B`	7, 17, 26	26
Qwen 3.5 4B base	`Qwen/Qwen3.5-4B-Base`	9, 22, 32	32
Large tier + current generation (single H200 141 GB)			
Gemma 4 26B-A4B base	`google/gemma-4-26B-A4B`	8, 18, 27	27
Mistral Small 24B base	`mistralai/Mistral-Small-24B-Base-2501`	10, 24, 37	37
Qwen 3.6 35B-A3B base	`Qwen/Qwen3.6-35B-A3B`	10, 24, 36	36
DeepSeek 67B base	`deepseek-ai/deepseek-llm-67b-base`	18, 48, 92	92
Catalan-aware control tier (BSC-LT, Apache 2.0)			
Salamandra 2B base	`BSC-LT/salamandra-2b`	6, 14, 21	21
Salamandra 7B base	`BSC-LT/salamandra-7b`	8, 19, 29	29
ALIA 40B base	`BSC-LT/ALIA-40b`	12, 29, 43	43

Table 3: The eleven-model base panel. The Salamandra/ALIA control tier — intentionally trained on a multilingual European corpus including a non-trivial Catalan share — is the indispensable causal control that separates the data hypothesis from the pipeline hypothesis.

Model	`anglo`	`bare`	`cat_fr`	`neu_ca`
ALIA-40b	0.504	0.490	0.457	0.483
deepseek-llm-67b	0.492	0.488	0.458	0.476
gemma2-2b	0.504	0.495	0.484	0.494
gemma4-26b-a4b	**0.435**	0.483	**0.370**	**0.403**
gemma4-e2b	0.458	0.476	0.435	0.448
mistral-small-24b	0.485	0.482	0.426	0.457
qwen2-1.5b	0.468	0.501	0.458	0.464
qwen3.5-4b-base	0.495	0.458	0.401	0.408
qwen3.6-35b-a3b	0.473	0.488	0.427	0.448
salamandra-2b	0.503	0.495	0.463	0.482
salamandra-7b	0.501	0.495	0.465	0.484

Table 4: Median `bias_ratio` pooled across ambiguity types (audited lexicon, n = 56). Values > 0.5 indicate pull toward the Anglo cluster; < 0.5, toward Catalan.

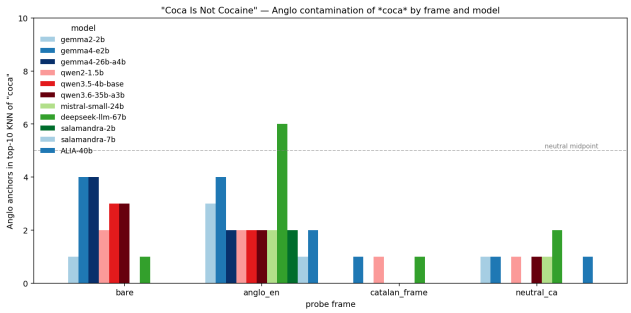


Figure 3: Anglo-anchor count (blue side of the binary `bias_ratio`) in the top-10 KNN of *coca* per (model, frame). Complement to Figure 4: together they tell which cluster catches *coca* when it drifts and which frame rescues it.

is a property of where *coca* lives in latent space, not induced by an English scaffold.

Under `catalan_frame` noise drops to 0 on three models (DeepSeek 67B, Gemma 4 E2B, Qwen 3.5 4B) and to 1 on Gemma 4 26B and Mistral Small 24B (only *Coca-Cola* survives); Gemma 4 26B's top-3 becomes *plaça, enxaneta, ajuntament*. The collision is rescuable, but only by a strong native frame, and the rescue depth is family-dependent: the Salamandra family and ALIA still carry 3–6 noise neighbours under `catalan_frame`, the cleanest evidence that intentional Catalan training does not, by itself, dissolve the **F1** collision.

7.4 The Catalan-aware control (Salamandra/ALIA)

The H6 verdict from the Catalan-aware extraction is that intentional Catalan training does not, by itself, erase the **F1**/**F2**/**F3** collisions. The collision modes survive across the Salamandra family up to ALIA-40B, suggesting they are properties of the **contextualized-embedding pipeline** (tokenization, mean-pooling, training-objective geometry) rather than of the data mixture. Three of four pre-specified H6 sub-hypotheses were falsified. *Without the Salamandra/ALIA panel we could not have distinguished the data-mix hypothesis from the pipeline hypothesis*: it is the indispensable causal control of this study.

7.5 The five-method statistical-rigour suite

Every model goes through five uniform additional analyses, all derivable from the committed artifacts without retraining: (1) per-concept paired-bootstrap CI on Δ (Figure 5, Table 6); (2) layer-depth × frame interaction (Figure 6a); (3) Anglo-subcluster pull on the focus concepts (Figure 6b); (4) stratified-bootstrap CIs on the model × ambiguity_type × frame median table (Figure 6c); and (5) Procrustes cross-model alignment (Figure 6d).

8 Discussion

Read the `domain_specific` (**F2**) stratum first: if *cercavila*, *correfoc*, *panellet* or *casteller* still sit closer to *giant / jester / enchanted forest* than to *festa major /*

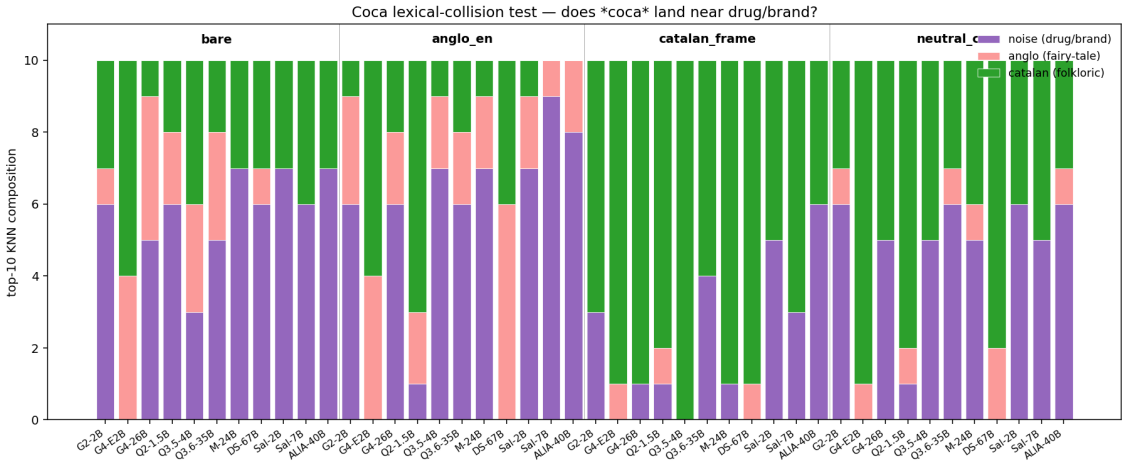


Figure 4: Purple = noise-cluster (drug / brand / beverage) hits in the top-10 KNN of *coca* per (model, frame). The pre-registered threshold $n_{\text{noise}} \geq 2$ outside `anglo_en` returns the strongly supported (structural) branch on ten of eleven models.

Model	bare	anglo_en	catalan_frame	neutral_ca
ALIA-40b	cocaine, Coca-Cola, caffeine	cocaine, Coca-Cola, soda	Coca-Cola, cocaine, soda	cocaine, Coca-Cola, soda
deepseek-llm-67b	Coca-Cola, coke, soda	enchanted forest, fairy, princess	cercavila, castellers, poble	enxaneta, plaça, grallers
gemma2-2b	Coca-Cola, coke, soda	Coca-Cola, coke, soda	Coca-Cola, coke, cobla	Coca-Cola, coke, cocaine
gemma4-26b-a4b	coke, caffeine, soda	Coca-Cola, cocaine, coke	plaça, enxaneta, ajuntament	Coca-Cola, cocaine, plaça
gemma4-e2b	cercavila, dwarf, dungeon	enchanted forest, peasant, pessebre	enxaneta, plaça, pessebre	pessebre, enxaneta, plaça
mistral-small-24b	coke, cocaine, Coca-Cola	Coca-Cola, coke, caffeine	plaça, Coca-Cola, pessebre	Coca-Cola, caffeine, coke
qwen2-1.5b	coke, soda, caffeine	cercavila, Coca-Cola, enxaneta	cercavila, enxaneta, Coca-Cola	cercavila, enxaneta, Coca-Cola
qwen3.5-4b-base	cercavila, soda, pilar	Coca-Cola, cocaine, coke	plaça, cobla, colla	plaça, Coca-Cola, diada
qwen3.6-35b-a3b	Coca-Cola, coke, soda	Coca-Cola, cocaine, coke	Coca-Cola, coke, cocaine	Coca-Cola, cocaine, coke
salamandra-2b	cocaine, Coca-Cola, caffeine	cocaine, Coca-Cola, caffeine	Coca-Cola, cocaine, cobla	cocaine, Coca-Cola, caffeine
salamandra-7b	cocaine, Coca-Cola, caffeine	cocaine, Coca-Cola, soda	Coca-Cola, cocaine, cobla	cocaine, Coca-Cola, caffeine

Table 5: The three nearest neighbours of *coca* per (model, frame). `catalan_frame` is the only frame that consistently rescues the concept toward the festive neighbourhood (*plaça, enxaneta, cercavila*).

Model	median Δ	n robust	n total
gemma2-2b	0.020	56	56
gemma4-e2b	0.024	49	50
gemma4-26b-a4b	0.068	56	56
qwen2-1.5b	0.010	56	56
qwen3.5-4b-base	0.095	56	56
qwen3.6-35b-a3b	0.045	56	56
mistral-small-24b	0.058	56	56
deepseek-llm-67b	0.034	56	56
salamandra-2b	0.039	56	56
salamandra-7b	0.035	56	56
ALIA-40b	0.046	56	56

Table 6: Median Δ per model and number of concepts whose 95 % CI excludes zero (n robust). All CIs are paired bootstrap. Qwen 3.5 4B has the highest median Δ (0.095): the most Catalan-frame-responsive. Qwen2 1.5B the lowest (0.010).

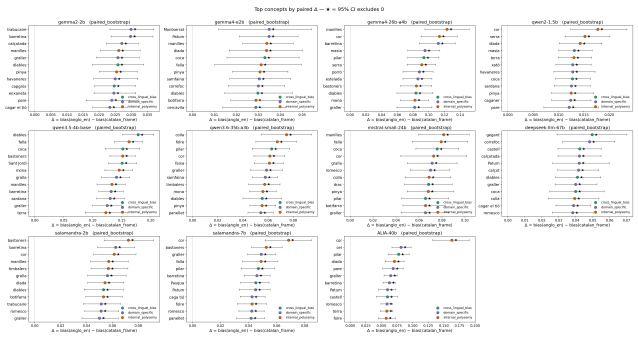


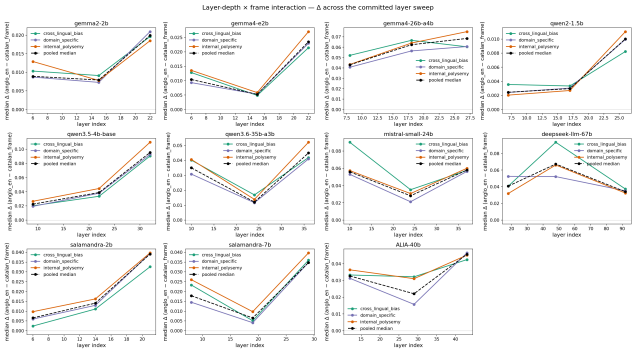
Figure 5: Forest plot of $\Delta(c) = \text{bias}(c, \text{anglo_en}) - \text{bias}(c, \text{catalan_frame})$ per concept with 95 % paired bootstrap CIs. Marked concepts have a CI excluding zero: robust evidence the Catalan frame changes the concept's geometry.

grallers / diables — even under a native Catalan frame — the pull is structural, not prompt-rescuable, and not explained by any lexical cognate.

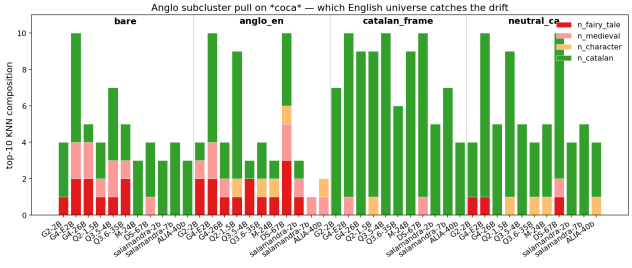
`cross_lingual_bias` (**F1**) items show the highest raw bias ratios, but that number is confounded with the cognate meaning of the Catalan word itself. They should not be read as the headline but as the *upper bound* of how strong the pull can get when the lexical shortcut is available.

`internal_polysemy` (**F3**) items are a different story: a high bias ratio on `bare` or `neutral_ca` most likely reflects the dominant mundane Catalan sense (pinecone, log, geological fault), not an English arrow. The frame differential is the interesting datum: if `catalan_frame` pulls the vector back into the festive neighbourhood, we were seeing polysemy; if not, even explicit Catalan festive context cannot override the default sense.

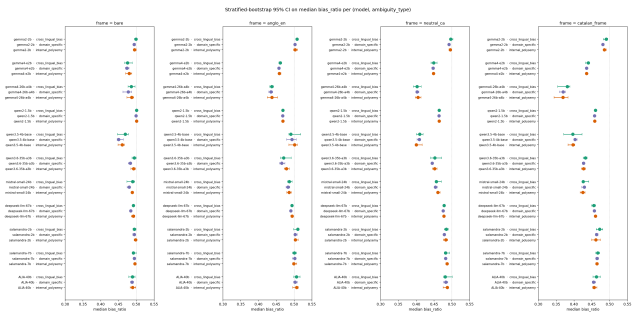
The `anglo_en` ↔ `catalan_frame` differential is there-



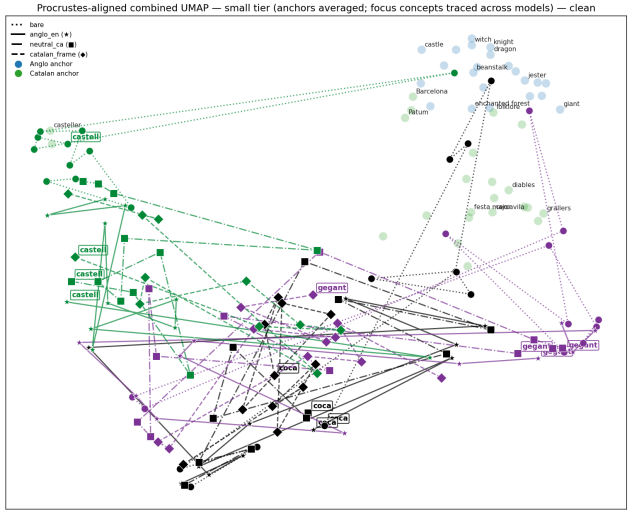
(a) Layer-depth × frame interaction.



(b) Anglo-subcluster pull on *coca*.



(c) Stratified-bootstrap CIs on the ambiguity-type medians.



(d) Procrustes alignment across the small-tier models.

Figure 6: The statistical-rigour suite. (a) whether the rescue happens early, mid or late in the stack; (b) which English universe (fairy-tale / medieval / character) catches *coca* when it drifts; (c) `cross_lingual_bias` has only 10 concepts, so its medians carry the widest CIs by design; (d) focus concepts traced as polylines across the chronologically-aligned small-tier models.

fore the key diagnostic: a large drop under `catalan_frame` means prompt-rescuable (the festive sense was retrievable, just not the default); no drop means the Catalan frame cannot override the English anchor — structural anglification.

9 Limitations

- **Mean-pooling over subwords** smears multi-token concepts (*pa amb tomàquet*, *caga tió*). A per-subword vector probe is queued for a future revision.
- **Hand-curated anchor lists** (22 / 23 / 13 entries): results are sensitive to that curation.
- **Cross-model raw cosines are not directly comparable.** The only surviving within-study comparisons are the normalized `bias_ratio`, the rank-based `top10_catalan_frac`, and the Procrustes alignment (small tier only).
- **Layer-selection circularity**, mitigated by reporting every metric across the three layers and by the per-concept paired-bootstrap CI.
- **Quantization is excluded a priori** (bf16 only): this is what excludes DeepSeek-V3 (671B) and V4-Flash (584B); the dense DeepSeek-LLM-67B-Base does fit on a single H200.
- **Base variants only** (no instruct); instruct comparisons would require a separate study.
- **One language.** The methodology is language-independent; a 15-concept replication in Occitan or Aragonese is queued.

What this study does not claim. That a Catalan-aware model is *worse* than a general-purpose one (the H6 verdict is that Catalan training does not cancel the inference-time requirement of a Catalan carrier prompt, not that Salamandra/ALIA is worse than Gemma); that the title-test result generalizes beyond *coca*; nor that the operational recommendation ("supply a Catalan carrier prompt") solves the deployment problem — it rescues retrieval-style probes but says nothing about generation quality, factual accuracy, or downstream tasks.

10 Conclusions

1. **The F1/F2/F3 taxonomy is necessary.** Under `anglo_en` on Qwen 3.5 4B the pooled median

`top10_catalan_frac` is 0.50 — a reader scanning a single number would conclude the model is unbiased. Stratifying reveals three stories inside that pool: **F1** = 0.55, **F2** = 0.50, **F3** = 0.40. The right intervention for **F3** is to teach the model that *pinya* has a second sense that is not a pinecone, not that *gegant* ≠ *giant*.

2. **The two families moved in opposite directions, and it survives at the large tier.** Gemma becomes context-invariant; Qwen becomes context-amplified. There is no global "best LLM for Catalan" — it depends on whether the deployment can guarantee Catalan framing.

3. **Scale is not the answer.** The biggest Qwen (3.6 35B-A3B) is less responsive to Catalan framing than the smaller Qwen 3.5 4B. ALIA-40B (Catalan-aware) has the deepest drop and the strongest native response: what matters is the multilingual pretraining diet *combined with* inference-time framing, not size.

4. **"Coca is not cocaine" is grounded by the metric, not the metaphor.** The pre-registered test ($n_{\text{noise}} \geq 2$ in the top-10 KNN of *coca*, outside `anglo_en`) returns the strongly supported (structural) branch on ten of eleven models. The sole exception is Gemma 4 E2B, which drifts into the Anglo fairy-tale cluster instead (**F2** catches it, the title test misses it). The **F1** reading is a literal property of latent space; the rescue is possible but only with a strong native frame, and its depth is family-dependent — the Catalan-aware control tier carries the heaviest residual noise.

Practical recommendation. The consequence is complementary to any pretraining strategy: **supply Catalan context in the inference-time prompt**, which works equally well for every model in the panel, general-purpose or Catalan-aware.

Data and code availability

The code (MIT), the hand-curated CSVs (CC-BY 4.0), all figures and extraction artifacts are public at `https://github.com/xaviviro/coca-is-not-cocaine`. The full methodological memo (in Catalan) is `metodologia.pdf` in the repository.

References

T. Bolukbasi, K.-W. Chang, J. Zou, V. Saligrama, A. Kalai. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. NeurIPS 2016. arXiv:1607.06520.

T. Dettmers, M. Lewis, Y. Belkada, L. Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. NeurIPS 2022. arXiv:2208.07339.

S. Li, B. Shafiq, P. Mathur, S. Roy. GeoMLAMA: Geographical and cultural biases in multilingual pre-trained language models. arXiv:2205.12247.

T. Naous, M. J. Ryan, A. Ritter, W. Xu. Having beer after prayer? Measuring cultural bias in large language models. ACL 2024. arXiv:2305.14456.

J. Rystrøm, H. R. Kirk, S. Hale. Multilingual ≠ multicultural. arXiv:2502.16534.

Y. Tao, O. Viberg, R. S. Baker, R. F. Kizilcec. Cultural bias and cultural alignment of large language models. PNAS Nexus 2024. arXiv:2311.14096.

W. Wang, M. Saar-Tsechansky, M. Sun, S. Wang. Not all countries celebrate Thanksgiving: On the cultural dominance in large language models. arXiv:2310.12481.

Ada Lovelace Institute. Tokenising culture: causes and consequences of cultural misalignment in LLMs. 2025. `https://www.adalovelaceinstitute.org/blog/cultural-misalignment-llms/`.